

National University of Computer and Emerging Sciences, Lahore Campus

	Course Name:	Computer Networks	Course Code:	CS 3001
	Program:	BS (Computer Science)	Semester:	Spring 2026
	Duration:	15 minutes	Total Marks:	15
	Paper Date:		Section	6B1
	Exam Type:	Quiz 5 - Chapter 5	Page(s):	2

Student Name

Roll No.

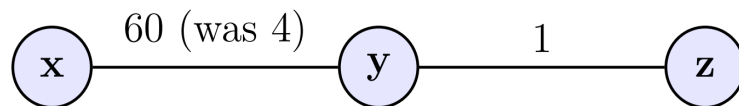
Section:

Q1. Carefully read the Question before attempting:

[5 marks]

Consider the 3-node linear topology below. Bellman-Ford Distance Vector routing is used. Link costs are: $c(x, y) = 4$, $c(y, z) = 1$. At t_0 , the network is stabilized. At t_1 , the link cost $c(x, y)$ dramatically increases to 60. Router y detects this change immediately. Router z does not see the link $x - y$ directly. Assume Poison Reverse is **NOT** enabled.

1. Calculate the Distance Vector entries $D_y(x)$ and $D_z(x)$ for steps t_1 , t_2 , t_3 .
2. Why does the network not stabilize (What is this problem called)? Give one way it can be prevented?



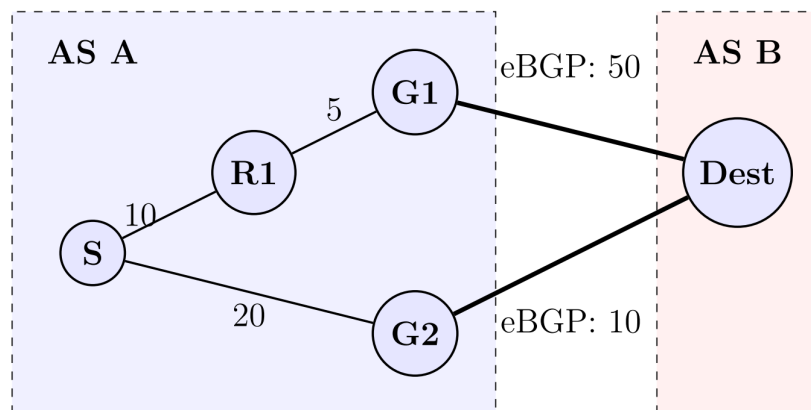
Q2. Carefully read the Question before attempting:

[5 marks]

AS A needs to send a packet to the destination IP in AS B. AS A has two gateways (G_1 , G_2) connected to AS B. The IGP (Intra-AS) costs within AS A are shown in the diagram. The eBGP costs (latency) on the links between AS A and AS B are also shown.

- **Hot Potato Routing:** Minimize intra-AS cost.
- **Cold Potato Routing:** Minimize total cost (Intra + Inter).

Calculate the path and total cost for the source node S for both routing strategies. What is the efficiency penalty (Total Cost Difference) of using Hot Potato routing in this specific topology?



Q3. Carefully read the Question before attempting:

[5 marks]

Compute the Forwarding Table for **Node E** (not the source A) using Dijkstra's algorithm. Note the directed edges and asymmetric costs. Run Dijkstra's algorithm originating at **Node E**. Provide the resulting Forwarding Table for Node E. (Format: Destination — Next Hop).

